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BRIEF COMMUNICATION

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Lactobacillus rhamnosus GG probiotic enteric regimen does not appreciably alter the gut microbiome or provide protection against GVHD after allogeneic hematopoietic stem cell transplantation

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Abstract

Graft-versus-host disease (GVHD) is a major adverse effect associated with allogeneic stem cell transplant. Previous studies in mice indicated that administration of the probiotic *Lactobacillus rhamnosus GG* can reduce the incidence of GVHD after hematopoietic stem cell transplant. Here we report results from the first randomized probiotic enteric regimen trial in which allogenic hematopoietic stem cell patients were supplemented with *Lactobacillus rhamnosus GG*. Gut microbiome analysis confirmed a previously reported gut microbiome association with GVHD. However, the clinical trial was terminated when interim analysis did not detect an appreciable probiotic-related change in the gut microbiome or incidence of GVHD. Additional studies are necessary to determine whether probiotics can alter the incidence of GVHD after allogeneic stem cell transplant.

KEYWORDS

graft-versus-host disease, gut microbiome, hematopoietic stem cell transplantation, probiotics

1 | INTRODUCTION

Graft-versus-host disease is a significant contributor of morbidity and mortality following allogeneic hematopoietic stem cell transplant. Recent evidence has suggested that GVHD may be associated with changes in the gut microbiota.[1] Alterations in the composition of the intestinal microbiota have been implicated in a variety of inflammatory and autoimmune disorders.[2,3] In allogeneic hematopoietic stem cell transplantation (HSCT), several studies have demonstrated an association between the gut microbiome diversity with outcomes such as survival and GVHD.[4] Allogeneic HSCT adversely impacts the commensal bacteria and reduces microbial diversity, leading to the domination of certain microbes.[4-6] Patients with lower intestinal diversity have been reported to have a 36% 3-year overall survival, whereas patients with intermediate and high intestinal diversity of microbiota had 60% and 67% overall 3-year survivals, respectively.[4] Gut microbiota trajectories pre-transplant and for the first 3-4 months following transplant have identified specific gut microbiota signatures associated with acute GVHD (aGVHD).[7] Moreover, decrease in health promoting species such as faecalibacterium, which promote short chain fatty acid

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production and mark microbiota-host mutualistic configuration, as well as increase in enterococci or streptococci have been suggested as specific dysbiosis seen in HSCT in GVHD.[5,8] Therefore, strategies to control alterations in intestinal microbiota are being evaluated.[9]

The supplementation of probiotics has demonstrated benefits in the setting of multiple medical conditions, particularly suggesting that probiotics may favorably impact gut-related immunity in humans.[10] Preliminary studies in animals have indicated that the administration of probiotics may modify GVHD following allogeneic hematopoietic stem cell transplant.[11] Gerbitz et al. observed that mice who received *Lactobacillus rhamnosus GG* prior to and following transplantation displayed lower concentrations of translocated microorganisms into mesenteric lymph nodes, reduced histologic inflammation and GVHD, and decreased post-transplantation mortality.

Here, we report the results from the first probiotic enteric regimen trial in which patients were supplemented with the probiotic, *Lactobacillus rhamnosus GG* (one capsule of 10 billion live *Lactobacillus rhamnosus GG* provided by Culturelle®).

2 | METHODS

This trial was approved by the Robert Wood Johnson Medical School Institutional Review Board. All eligible patients who underwent allogeneic stem cell transplantation at the Rutgers Cancer Institute of New Jersey (RCINJ)—Robert Wood Johnson University Hospital—were offered participation, excluding those with prior use of probiotic supplementation within the last 90 days, or those with a history of inflammatory bowel disease. Data for these patients were abstracted from electronic medical records on file at RCINJ's Office of Human Research Services (OHRS) and from data forms submitted to the Center for International Blood and Marrow Transplant Registry (CIBMTR). Patients receiving probiotics recorded study medicine ingestion in a food diary.

To longitudinally profile the microbiome makeup, we performed 16S sequencing of samples from five control patients and nine who received probiotics, prior to the start of the trial and at the first and third month intervals. DNA was extracted from stool samples, followed by Illumina MiSeq sequencing and classification of the Operational Taxonomic Units (OTUs) at family and genus levels, using Greengenes reference database, as previously described.[7] We assessed differential microbial population among samples using DeSeq2, as implemented in the Qiime pipeline and compared microbial raw counts across various biological states.[12-14] To include information on sequential samples from patients who developed GVHD vs those who did not, we used a Zero-Inflated Beta Random Effect model (ZIBR).[15] We performed these analyses at both family and genus levels, excluding OTUs without annotations and limiting to those with abundances higher than 0.001%.

3 | RESULTS

First, we demonstrated that probiotic supplementation (one capsule of 10 billion live *Lactobacillus rhamnosus GG* provided by Culturelle®) was safe and did not lead to unexpected adverse events or attributable infections compared to historic control patients. We then performed a trial in which we randomized 31 patients in a 2:1 ratio; 20 received *Lactobacillus rhamnosus GG* administration, and 11 did not receive any probiotics. Treatment started at time of engraftment. The end points of this analysis were the incidence of GVHD, and the impact of therapy on the composition of the gut microbiome post-transplant.

Among the 20 patients who were treated with probiotics, at the 3-month interval, three patients developed grade I aGVHD, six experienced grade II aGVHD, and 11 were free of disease. At 6 months, three patients had limited cGVHD, three developed extensive cGVHD, nine were without disease, one underwent a second transplant, and four patients expired. At 12 months, three patients had limited cGVHD, five had extensive cGVHD, five were without disease, five had expired, one underwent a second transplantation, and one had yet to reach the 12-month interval at the time of these results (Table 1).

For the 11 randomized control patients, at the 3-month interval, one patient developed grade I aGVHD, three had grade II aGVHD, and seven were without disease. At 6 months, one patient had limited cGVHD, one had extensive cGVHD, six were without disease, and three had expired. At 12 months, one patient had limited cGVHD, one had extensive cGVHD, four had no evidence of disease, three had expired, and two had yet to reach the interval at the time of these results (Table 1). We found no significant statistical evidence for differential development of GVHD between the two cohorts (Fisher's exact test P=.38).

Our results indicated also that patients maintained their respective initial microbial diversity during the course of monitoring, independent of treatment, as measured by Shannon index (Figure 1A), as well as Inverse Simpsons index (data not shown). Although patients who received probiotics showed slightly higher microbial diversity prior to receiving treatment compared to those who did not (Wilcoxon rank-sum test P<.05), differential abundance analysis did not identify statistically significant changes in microbial populations during the course of monitoring. In particular, *Lactobacillaceae* family and *Lactobacillus* genus abundances were not affected by probiotic administration. The study was terminated after preliminary analysis indicated lack of appreciable effect on the gut microbiome or incidence of GVHD.

4 | DISCUSSION

Our findings suggest that supplementing therapy with *Lactobacillus rhamnosus GG* did not significantly modify the gut microbiome diversity and, unlike results in murine studies, did not impact the incidence or severity of GVHD. However, we noted that among the randomized control patients, the abundance of the genus *Blautia* of the family *Lachnospiraceae* was significantly higher among the patients who did not develop GVHD as compared with those patients who developed GVHD at the 3-month interval. Similar results were found for the genus *Ruminococcus* of the family *Lachnospiraceae* (Figure 1B). These results corroborate prior findings that reported increased concentrations of the genus *Blautia* to be associated with reduced GVHD

TABLE 1 Patient characteristics

Patient ID	Age	Diagnosis	Type	ARM	Month 3 aGVHD	Month 3 cGVHD	Month 6 aGVHD	Month 6 cGVHD	Month 12 aGVHD	Month 12 cGVHD
002	55	MM	MRD	No Probiotic	No	No	No	Yes	No	Yes
004	68	AML	MUD	No Probiotic	No	No	Yes	No	Yes	No
008	51	PTCL	MUD	No Probiotic	No	No	No	No	No	No
011	63	Angioimmunoblastic T cell NHL	MUD	No Probiotic	Yes	No	Yes	No	No	Yes
014	35	AML	MUD	No Probiotic	Yes	No	Yes	Pt. Expired	Pt. Expired	Pt. Expired
016	56	AML	MUD	No Probiotic	No	No	Pt. Expired	Pt. Expired	Pt. Expired	Pt. Expired
021	53	Pre B ALL	MRD	No Probiotic	Yes	No	Pt. Expired	Pt. Expired	Pt. Expired	Pt. Expired
024	68	DLBCL/NHL	Haplo	No Probiotic	Yes	No	No	No	No	No
025	64	AML	MUD	No Probiotic	No	No	No	No	No	No
030	40	AML	MUD	No Probiotic	No	No	No	Yes	N/A	N/A
032	63	Phil neg. ALL	Haplo	No Probiotic	No	No	No	No	N/A	N/A
003	47	MM	MRD	Probiotic	No	No	No	No	No	Yes
005	28	HTLV-1 Lymphoma/NHL	MUD	Probiotic	Yes	No	2nd Transplant	2nd Transplant	2nd Transplant	2nd Transplant
006	65	AML	MUD	Probiotic	No	No	No	No	No	Yes
007	47	AML	MUD	Probiotic	Yes	No	Pt. Expired	Pt. Expired	Pt. Expired	Pt. Expired
009	48	AML	MUD	Probiotic	Yes	No	No	Yes	Pt. Expired	Pt. Expired
010	63	AML	MUD	Probiotic	Yes	No	Yes	Pt. Expired	Pt. Expired	Pt. Expired
012	25	AML	MUD	Probiotic	No	No	No	No	No	No
013	42	AML	MUD	Probiotic	Yes	No	No	Yes	No	No
015	63	AML	MUD	Probiotic	No	No	No	Pt. Expired	Pt. Expired	Pt. Expired
017	46	NHL (follicular)	MUD	Probiotic	No	No	No	No	No	No
018	55	MDS	MUD	Probiotic	No	No	No	Yes	No	Yes
019	43	AML	MUD	Probiotic	No	No	No	No	No	Yes
020	23	ALL	MUD	Probiotic	Yes	No	No	Yes	No	Yes
022	58	Ph+ ALL	MRD	Probiotic	Yes	No	No	Yes	No	Yes
023	56	Aplastic Anemia	MUD	Probiotic	No	No	No	No	No	No
026	59	MM	MUD	Probiotic	Yes	No	Yes	No	No	Yes
027	55	Ph + ALL	MUD	Probiotic	No	No	No	No	No	No
028	59	AML	MUD	Probiotic	No	No	No	No	No	Yes
029	66	MDS w/underlying CML	MUD	Probiotic	Yes	No	No	Yes	N/A	N/A
031	48	AML	MUD	Probiotic	No	Pt. Expired	Pt. Expired	Pt. Expired	Pt. Expired	Pt. Expired

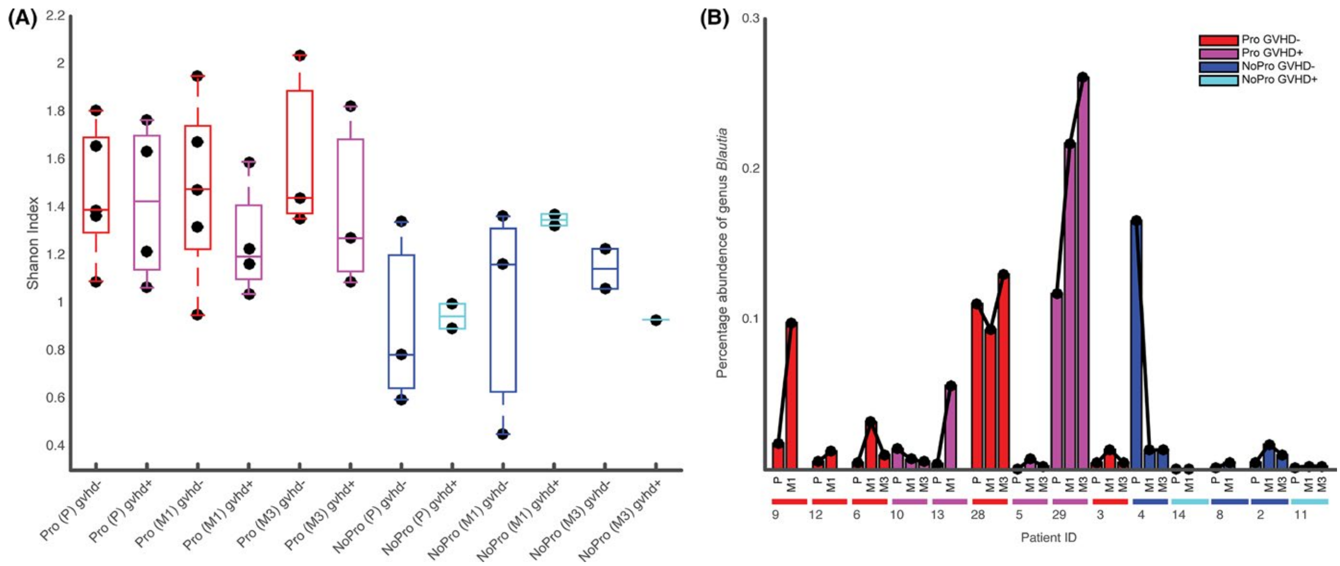

FIGURE 1 Diversity and microbial abundance in longitudinal samples. (A) Shannon index shows maintenance of initial microbial diversity during the course of monitoring, independent of treatment. (B) In control patients, the abundance of the genus *Blautia* was found significantly higher among the patients who did not develop GVHD compared to those who did at the 3-mo interval. This or another genus was not associated with the development of GVHD in patients who received probiotics. P, M1, M3 indicate samples collected prior start of the trial, at the 1- and 3-mo intervals

lethality in allogeneic stem cell transplantation patients.[7] In patients who received probiotics, however, the abundances of neither *Blautia* nor *Ruminococcus* were associated with the development of GVHD.

Without any evidence that probiotic therapy induced a change in the gut microbiome, the study was terminated. Despite the relatively small sample size of this study, and slightly higher baseline microbial diversity among patients who subsequently received probiotics, our results suggest the potentially significant implications of gut microbiome in the development of GVHD for allogeneic stem cell transplant patients. Further investigations are required to understand how probiotics may lead to elimination of protective impact of some bacteria against GVHD. Additional studies with different probiotics preparations starting earlier in the transplant process are planned.

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CONFLICT OF INTEREST

None.

AUTHORS' CONTRIBUTIONS

Elan Gorshein, and Catherine Wei: Drafting article, revision of article; Susan Ambrosy, Shanna Budney, and Juliana Vivas: Data collection; Angelica Shenkerman, Jacqueline Manago, MaryKate McGrath, and Anne Tyno: Concept/design; Yong Lin: Statistics; Vimal Patel, Mecide Gharibo, and Dale Schaar: Concept/design; Robert Jenq: Concept/design, gut microbiome analysis; Hossein Khiabanian: Data analysis, statistics, data interpretation; Roger Strair: Concept/design, critical revision of article.

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